Freda-Marie Beecham

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EDUCATION

Ashesi University **Berekuso, Eastern Region**
BSc. Computer Science | GPA: 3.78 **Jan 2022 – Jun 2025**
Relevant Courses: Internet of Things, Data Structures and Algorithms, Networks and Data Communication, Data Science

ACHIEVEMENTS/AWARDS

3rd Place, Nestlé Ghana Live Business Case **Nov 2024**
Dean's List Honors (GPA of 3.5 and above), Ashesi University **Jan 2022 – Dec 2024**

WORK EXPERIENCE

Wahu Mobility – Accra, Ghana **Nov 2025 – Present**
Software Engineer

- Developed the Wahu OS, an internal operations platform to centralize workflows across 4 business pillars, integrating dashboards for fleet management, production, rider productivity, and billing to improve operational visibility.
- Automated maintenance logging for 826 vehicles by developing a Google Apps Script workflow that tracks vehicle entry into and exit from maintenance, calculates the time out of service and records completion status.
- Standardised and migrated 1,022 rider records across 5 Excel spreadsheets into Wahu OS, resolving inconsistencies and missing information to improve operational data consistency.
- Integrated a telematics API for real-time vehicle telemetry and IoT device communication, enabling accurate location tracking and data ingestion into the fleet management system.
- Built the Wahu Client Portal using TypeScript and C#, enabling B2B clients to remotely monitor electric vehicles in real-time.
- Integrated Flespi API for real-time vehicle telemetry and IoT device communication, enabling accurate location tracking and data ingestion into the fleet management system.
- Designed and deployed backend services on Google Cloud Platform (GCP), leveraging Cloud Run for containerized microservices and Cloud Storage for asset management.
- Monitored application health and debugged production issues using GCP logging and observability tools.
- Managed Docker-based deployments with environment configurations for Elasticsearch, Kafka, and Redis, maintaining service reliability across distributed infrastructure.
- Managed feature branches, conducted code reviews and executed production releases for NET solution projects.
- Researched database design principles and optimization strategies to guide the architecture of a new in-house fleet management system.

AmaliTech Services GmbH - Accra, Ghana **Jun 2023 – Aug 2023**
Backend Developer Intern

- Implemented PA Authentication functions as a member of a collaborative Scrum team for a hospital booking management system, using Python Django REST Framework, in a 6-week span
- Collaborated on the full-stack design and development of an Access Key Manager API, implementing secure key management functionalities, ensuring safe and reliable access management for users

Ondio Services **Sept 2022 – May 2023**
Video Editor

- Edited 6 videos, for the Stories from the Motherland series to engage 77,000 subscribers
- Coordinated training sessions, teaching color correction, audio mixing and motion graphics development in Adobe Premiere Pro

Seatec Telecom Services Limited - Tema **Aug 2022**
Solutions Design Intern

- Evaluated 5 conferencing products against client-specific criteria to propose optimal IT solutions,
- recommending a secure, user-focused solution for the client
- Modified existing company footage to create endorsement and advertisement videos, enhancing proposal presentations

PROJECTS AND RESEARCH

Applied Capstone Project **Sep 2024 – May 2025**
Enhancing Ride-Hailing Passenger Safety and Comfort for Women in Ghana Using AI

- Conducted user surveys and prototype testing over 2 months to gather insights on safety concerns and evaluate the usability of AI-powered features designed to protect female passengers
- Built a mobile safety app using on-device AI to detect early warning signs of danger during rides through geofencing, audio analysis, and facial recognition, with offline alerts for low-connectivity environments

IoT-Based Smart Environment Monitoring & Control System **Dec 2024**

- Deployed a smart sensor network using ESP32-WROOM modules to monitor temperature, humidity, and light intensity in indoor and outdoor environments, enabling real-time environmental tracking
- Integrated machine learning to predict future temperature trends based on location, LDR values, humidity, and time of day, allowing for enhanced energy efficiency

Ashesi University *JEC Management System **Jul 2024 - Aug 2024**

- Developed a React-based attendance tracker, scheduler, and voting portal, as part of a 4-member team, to improve transparency in university elections

Emotion Recognition in Ghanaian Dialects **Jul 2024 – Aug 2024**

- Built a deep learning model to identify emotions in speech, using Tensorflow, to enhance the effectiveness of emotion recognition in 5 local dialects

Sankofa GirlsInScience Programme Website **May 2024**

- Enhanced the outreach and impact of the Sankofa GirlsInScience Programme by designing and developing a website to amplify their message and expand their reach

Sayf Crowdsourced Review Platform **Feb 2024 - Apr 2024**

- Designed and developed a data-powered web application to enhance passenger safety in Ghana's ride-hailing services by enabling user reviews, incident reporting, driver flagging, and a community chat forum

Envision 7 Online Learning Platform

- Collaborated, within a team of 9 students, to develop an online learning platform, using Squarespace and WordPress, to provide Ghanaian Junior High School students with IT skills

Research on the Impact of Online Learning on Ghanaian Students **Sep 2020 – Feb 2021**

- Conducted 2-month long quantitative and qualitative research to analyze the extent to which online learning has transformed the educational experiences of Ghanaian high school students in the Greater Accra Region

LEADERSHIP AND ACTIVITIES

Ashesi Student Council Academic Committee, Ashesi University **Jan 2024 – Dec 2024**

Head of the Course Representative System

- Developed a comprehensive Excel-based feedback system to monitor course representative reports
- Managed a team of 15 students to gather feedback from the student body, facilitating effective decision making and improvements in course designs

Academic Peer Tutoring Program, Ashesi University **Feb 2023 – Dec 2024**

Programming Peer Tutor

- Led 10 sessions on Data Structures and Algorithms, covering key concepts such as arrays, linked lists, and trees, which significantly improved students' understanding and application of these concepts
- Aided 30 students in mastering the concepts of inheritance, polymorphism and encapsulation in Java through targeted instruction and hands-on practice on a weekly basis

Peer Tutoring Program, Ashesi University **Jan 2023 - May 2023**

Peer Tutor

- Enhanced students' narrative and persuasive composition skills through detailed review sessions
- Aided students in mastering the concepts of inheritance, polymorphism and encapsulation in Java, through targeted instruction and hands-on practice

SKILLS

- Programming Languages: C#, C++, CSS, HTML, Java, JavaScript, Kotlin, PHP, Python, SQL, R
- Databases: PostgreSQL, MySQL, Elasticsearch
- Cloud and DevOps: Google Cloud Platform (Cloud Run, Cloud Storage, GCP Logging), Docker
- Frameworks & Libraries: Android Studio, Django Rest Framework, Flutter, React.js, Tensorflow
- Exceptionally skilled in Adobe Premiere Pro, Photoshop, Illustrator, Squarespace, Wix and Canva

REFERENCES

Available upon request